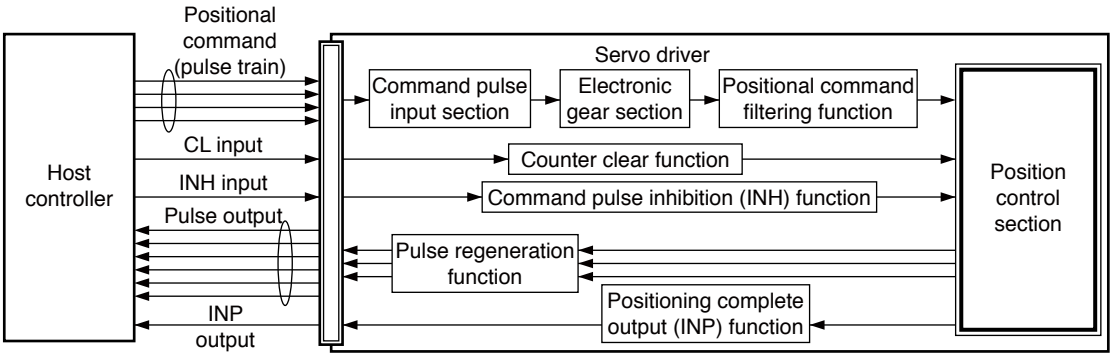
1. Outline of mode

Position Control Mode

Outline

You can perform position control based on the positional command (pulse train) from the host controller.

This section describes the fundamental setup to be used for the position control.



Function

(1) Process of command pulse input

The positional commands of the following 3 types (pulse train) are available.

- 2-phase pulse
- Positive direction pulse/negative direction pulse
- Pulse train + sign

Set the pulse configuration and pulse counting method based on the specification and configuration of installation of the host controller.

The input terminals can accommodate the following 2 systems.

- Input 1 “PULSH1, PULSH2, SIGNH1, SIGNH2” line receiver input (4 Mpps)
- Input 2 “PULS1, PULS2, SIGN1, SIGN2” photocoupler input (500 kpps)

Caution For line driver output, “Input 2” can also be used without changing the allowable input frequency.

- **Relevant parameters**

Parameter No.	Title	Range	Function
Pr0.05	Selection of command pulse input	0 to 1	You can select either the photo-coupler input or the exclusive input for line driver as the command pulse input.
Pr0.06	Command pulse rotational direction setup	0 to 1	Sets the counting direction when command pulse is input.
Pr0.07	Command pulse input mode setup	0 to 3	Sets the counting method when command pulse is input.

Note For details of these parameters, refer to P.4-6 and 7 “Details of parameter”.

Related page
- P.3-14 “Control Block Diagram”
- P.3-18 “Wiring Diagram to the connector, X4”

(2) Electronic gear function

This function multiplies the input pulse command from the host controller by the predetermined dividing or multiplying factor and applies the result to the position control section as the positional command. By using this function, desired motor rotations or movement distance per unit input command pulse can be set; or the command pulse frequency can be increased if the desired motor speed cannot be obtained due to limited pulse output capacity of the host controller.

• Relevant parameters

Parameter No.	Title	Range	Function
Pr0.08	Command pulse counts per one motor revolution	0 to 1048576	Set the command pulses that causes single turn of the motor shaft.
Pr0.09	1st numerator of electronic gear	0 to 1073741824	Set the numerator of division/multiplication operation made according to the command pulse input.
Pr0.10	Denominator of electronic gear	1 to 1073741824	Set the Denominator of division/multiplication operation made according to the command pulse input.

Note For details of these parameters, refer to P.4-8 “Details of parameter”.

(3) Positional command filtering function

To make the positional command divided or multiplied by the electronic gear smooth, set the command filter.

• Relevant parameters

Parameter No.	Title	Range	Unit	Function
Pr2.22	Positional command smoothing filter	0 to 10000	0.1ms	Set up the time constant of the 1st delay filter in response to the positional command.
Pr2.23	Positional command FIR filter	0 to 10000	0.1ms	Set up the time constant of the 1st delay filter in response to the positional command.

Note For details of these parameters, refer to P.4-22 and 23 “Details of parameter”.

(4) Pulse regeneration function

The information on the amount of movement can be sent to the host controller in the form of A- and B-phase pulses from the servo driver. When the output source is the encoder, Z-phase signal is output once per motor revolution; or if the feedback scale, the signal is output at absolute zero position. The output resolution, B-phase logic and output source (encoder or feedback scale) can be set with parameters.

• Relevant parameters

Parameter No.	Title	Range	Unit	Function
Pr0.11	Output pulse counts per one motor revolution	1 to 262144	P/r	You can set up the output pulse counts per one motor revolution for each OA and OB with the Pr0.11 setup.
Pr0.12	Reversal of pulse output logic	0 to 3	—	You can set up the B-phase logic and the output source of the pulse output. With this parameter, you can reverse the phase relation between the A-phase pulse and the B-phase pulse by reversing the B-phase logic.
Pr5.03	Denominator of pulse output division	0 to 262144	—	For application where the number of pulses per revolution is not an integer, this parameter can be set to a value other than 0, and the dividing ratio can be set by setting numerator of division to Pr0.11 and denominator of division to Pr5.03.
Pr5.33	Pulse regenerative output limit setup	0 to 1	—	Enable/disable detection of Err28.0 Pulse regenerative limit protection.
Pr6.20	Z-phase setup of external scale	0 to 400	µs	Set up the Z phase regenerative width of feedback scale in unit of time.
Pr6.21	Serial absolute external scale Z phase setup	0 to 268435456	pulse	Full-closed control using serial absolute feedback scale. When outputting pulses by using the feedback scale as the source of the output, set the Z phase output interval in units of A phase output pulses of the feedback scale (before multiplied by 4).
Pr6.22	A, B phase external scale pulse output method selection	0 to 1	—	Select the pulse regeneration method of A, B and Z parallel feedback scale.

Note For details of these parameters, refer to P.4-8, 9, 42, 49 and 54 “Details of parameter”.

(5) Deviation counter clear function

The deviation counter clear input (CL) clears the counts of positional deviation counter at the position control to 0.

• Relevant parameters

Parameter No.	Title	Range	Function
Pr5.17	Counter clear input mode	0 to 4	You can set up the clearing conditions of the counter clear input signal.

Note For details of these parameters, refer to P.4-46 “Details of parameter”.

(6) Positioning complete output (INP) function

The completion of positioning can be verified by the positioning complete output (INP). When the absolute value of the positional deviation counter at the position control is equal to or below the positioning complete range by the parameter, the output is ON. Presence and absence of positional command can be specified as one of judgment conditions.

• Relevant parameters

Parameter No.	Title	Range	Unit	Function
Pr4.31	Positioning complete (In-position) range	0 to 262144	Command unit	Set up the timing of positional deviation at which the positioning complete signal (INP1) is output.
Pr4.32	Positioning complete (In-position) output setup	0 to 3	—	Select the condition to output the positioning complete signal (INP1).
Pr4.33	INP hold time	0 to 30000	1ms	Set up the hold time when Pr4.32 Positioning complete output setup = 13.
Pr4.42	2nd Positioning complete (In-position) range	0 to 262144	Command unit	Set up the timing of positional deviation at which the positioning complete signal (INP2) is output.

Note For details of these parameters, refer to P.4-38 and 41 “Details of parameter”.

(7) Command pulse inhibition (INH) function

The command pulse input counting process can be forcibly terminated by using the command pulse inhibit input signal (INH). When INH input is ON, the servo driver ignores the command pulse, disabling pulse counting function.

The default setting of this inhibition function is disable. To use INH function, change the setting of Pr5.18 “Invalidation of command pulse prohibition input”.

• Relevant parameters

Parameter No.	Title	Range	Function
Pr5.18	Invalidation of command pulse inhibit input	0 to 1	Select command pulse inhibit input enable/disable.
Pr5.19	Command pulse inhibit input reading setup	0 to 4	Select command pulse inhibit input enable/disable signal reading period. When the status of several signals read during the predetermined reading period are same, update the signal status.

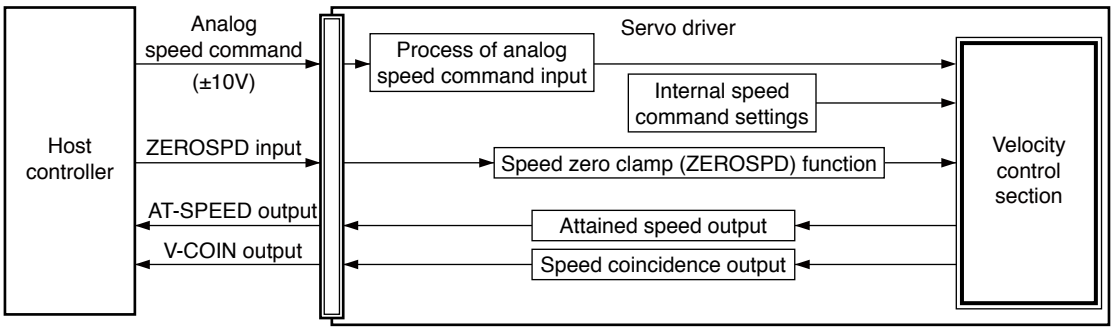
Note For details of these parameters, refer to P.4-46 “Details of parameter”.

1. Outline of mode

Velocity Control Mode

Outline

You can control the speed according to the analog speed command from the host controller or the speed command set in the servo driver.



Function

(1) Velocity control by analog speed command

The analog speed command input voltage is converted to equivalent digital speed command. You can set the filter to eliminate noise or adjust the offset.

• Relevant parameters

Parameter No.	Title	Range	Unit	Function
Pr3.00	Speed setup, Internal/ External switching	0 to 3	—	This driver is equipped with internal speed setup function so that you can control the speed with contact inputs only.
Pr3.01	Speed command rotational direction selection	0 to 1	—	Select the Positive/Negative direction specifying method.
Pr3.02	Input gain of speed command	10 to 2000	(r/min) /V	Based on the voltage applied to the analog speed command (SPR), set up the conversion gain to motor command speed.
Pr3.03	Reversal of speed command input	0 to 1	—	Specify the polarity of the voltage applied to the analog speed command (SPR).
Pr4.22	Analog input 1 (AI1) offset setup	–5578 to 5578	0.359mV	Set up the offset correction value applied to the voltage fed to the analog input 1.
Pr4.23	Analog input 1 (AI1) filter	0 to 6400	0.01ms	Set up the time constant of 1st delay filter that determines the lag time behind the voltage applied to the analog input 1.

Note For details of these parameters, refer to P.4-24, 25 and 37 “Details of parameter”.

Related page
• P.3-14 “Control Block Diagram”
• P.3-18 “Wiring Diagram to the connector, X4”

(2) Velocity control by internal speed command

You can control the speed by using the internal speed command set to the parameter. By using the internal speed command selections 1, 2, 3 (INTSPD1, 2, 3), you can select best appropriate one among up to 8 internal speed command settings. Default setting uses the analog speed command. To use the internal speed command, select it through Pr3.00 “Internal/external speed setup”.

• Relevant parameters

Parameter No.	Title	Range	Unit	Function
Pr3.00	Speed setup, Internal/ External switching	0 to 3	—	This driver is equipped with internal speed setup function so that you can control the speed with contact inputs only.
Pr3.01	Speed command rotational direction selection	0 to 1	—	Select the Positive/Negative direction specifying method.
Pr3.04	1st speed of speed setup	−20000 to 20000	r/min	Set up internal command speeds, 1st to 1st.
Pr3.05	2nd speed of speed setup			Set up internal command speeds, 1st to 2nd.
Pr3.06	3rd speed of speed setup			Set up internal command speeds, 1st to 3rd.
Pr3.07	4th speed of speed setup			Set up internal command speeds, 1st to 4th.
Pr3.08	5th speed of speed setup			Set up internal command speeds, 1st to 5th.
Pr3.09	6th speed of speed setup			Set up internal command speeds, 1st to 6th.
Pr3.10	7th speed of speed setup			Set up internal command speeds, 1st to 7th.
Pr3.11	8th speed of speed setup			Set up internal command speeds, 1st to 8th.

Note For details of these parameters, refer to P.4-24, 25 and 26 “Details of parameter”.

(3) Speed zero clamp (ZEROSPD) function

You can forcibly set the speed command to 0 by using the speed zero clamp input.

• Relevant parameters

Parameter No.	Title	Range	Unit	Function
Pr3.15	Speed zero-clamp function selection	0 to 3	—	You can set up the function of the speed zero clamp input.
Pr3.16	Speed zero clamp level	0 to 1	r/min	elect the timing at which the position control is activated as the Pr3.15 Speed zero-clamp function selection is set to 2 or 3.

Note For details of these parameters, refer to P.4-27 “Details of parameter”.

(4) Attained speed output (AT-SPEED)

The signal AT-SPEED is output as the motor reaches the speed set to Pr4.36 “Attained speed”.

• Relevant parameters

Parameter No.	Title	Range	Unit	Function
Pr4.36	At-speed (Speed arrival)	10 to 20000	r/min	Set the detection timing of the speed arrival output (AT-SPEED).

Note For details of these parameters, refer to P.4-40 “Details of parameter”.

(5) Speed coincidence output (V-COIN)

This signal is output when the motor speed is equal to the speed specified by the speed command. The motor speed is judged to be coincident with the specified speed when the difference from the speed command before/after acceleration/deceleration is within the range specified by Pr4.35 “Speed coincident range” .

• Relevant parameters

Parameter No.	Title	Range	Unit	Function
Pr4.35	Speed coincidence range	10 to 20000	r/min	Set the speed coincidence (V-COIN) output detection timing.

Note For details of these parameters, refer to P.4-39 “Details of parameter”.

(6) Speed command acceleration/deceleration setting function

This function controls the speed by adding acceleration or deceleration instruction in the driver to the input speed command.

Using this function, you can use the soft start when inputting stepwise speed command or when using internal speed setup. You can also use S shaped acceleration/deceleration function to minimize shock due to change in speed.

• Relevant parameters

Parameter No.	Title	Range	Unit	Function
Pr3.12	Acceleration time setup	0 to 10000	ms/ (1000r/min)	Set up acceleration processing time in response to the speed command input.
Pr3.13	Deceleration time setup	0 to 10000	ms/ (1000r/min)	Set up deceleration processing time in response to the speed command input.
Pr3.14	Sigmoid acceleration/ deceleration time setup	0 to 1000	ms	Set S-curve time for acceleration/ deceleration process when the speed command is applied.

Caution When the position loop is external to the driver, do not use the acceleration/ deceleration time setting. Set these values to 0.

Note For details of these parameters, refer to P.4-26 and 27 “Details of parameter”.

1. Outline of mode

Torque Control Mode

Outline

The torque control is performed according to the torque command specified in the form of analog voltage. For controlling the torque, the speed limit input is required in addition to the torque command to maintain the motor speed within the speed limit.

With the A5 series, 3 torque control modes are available, each requires different torque command and speed limit as shown in the table below.

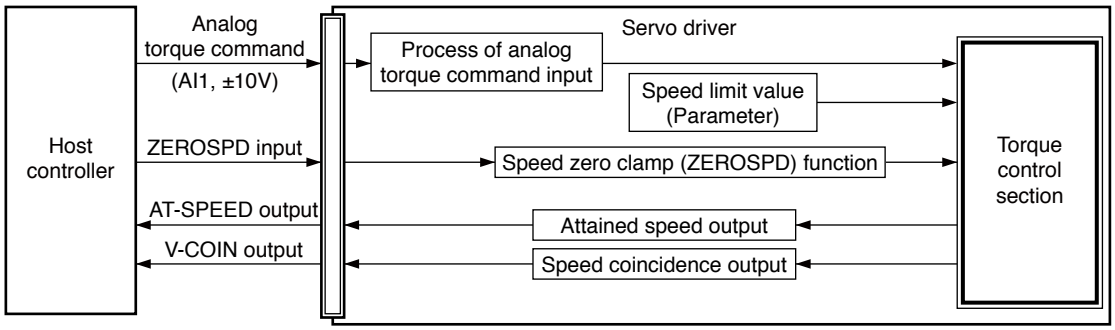
• Pr3.17 (Selection of torque command)

Setup value		Torque command input	Velocity limit input
0	Selection of torque command 1	Analog input 1 *1 (AI1, 16-bit resolution)	Parameter value (Pr3.25)
1	Selection of torque command 2	Analog input 2 (AI2, 12-bit resolution)	Analog input 1 (AI1, 16-bit resolution)
2	Selection of torque command 3	Analog input 1 *1 (AI1, 16-bit resolution)	Parameter value (Pr3.25, Pr3.26)

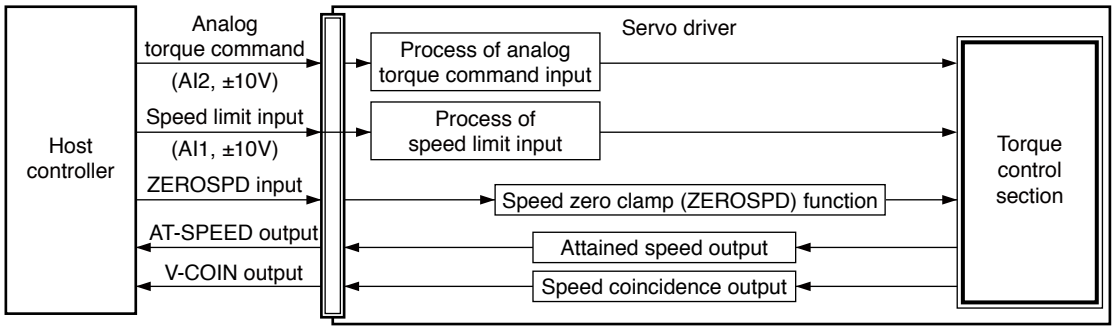
*1 For Pr0.01 Control mode setup = 5 (velocity/torque control), the torque command input is the analog input 2 (AI2, 12-bit resolution).

Note For details of these parameters, refer to P.4-28 “Details of parameter”.

<Selection of torque command 1, 3>



<Selection of torque command2>



Related page
- P.3-16 “Control Block Diagram”
- P.3-19 “Wiring Diagram to the connector, X4”

Function

(1) Process of analog torque command input

This process converts the analog torque command input (voltage) to the equivalent digital torque command having the same effect. You can set the filter or adjust the offset to eliminate noise.

• Relevant parameters <Selection of torque command 1, 3>

Parameter No.	Title	Range	Unit	Function
Pr3.18	Torque command direction selection	0 to 1	—	Select the direction positive/negative direction of torque command.
Pr3.19	Input gain of torque command	10 to 100	0.1V /100%	Based on the voltage (V) applied to the analog torque command (TRQ R), set up the conversion gain to torque command (%).
Pr3.20	Input reversal of torque command	0 to 1	—	Set up the polarity of the voltage applied to the analog torque command (TRQR).
Pr4.22	Analog input 1 (AI1) offset setup	–5578 to 5578	0.359mV	Set up the offset correction value applied to the voltage fed to the analog input 1.
Pr4.23	Analog input 1 (AI1) filter	0 to 6400	0.01ms	Set up the time constant of 1st delay filter that determines the lag time behind the voltage applied to the analog input 1.

• Relevant parameters <Selection of torque command 2>

Parameter No.	Title	Range	Unit	Function
Pr3.18	Torque command direction selection	0 to 1	—	Select the direction positive/negative direction of torque command.
Pr3.19	Input gain of torque command	10 to 100	0.1V /100%	Based on the voltage (V) applied to the analog torque command (TRQ R), set up the conversion gain to torque command (%).
Pr3.20	Input reversal of torque command	0 to 1	—	Set up the polarity of the voltage applied to the analog torque command (TRQR).
Pr4.25	Analog input 2 (AI2) offset setup	–342 to 342	5.86mV	Set up the offset correction value applied to the voltage fed to the analog input 2.
Pr4.26	Analog input 2 (AI2) filter	0 to 6400	0.01ms	Set up the time constant of 1st delay filter that determines the lag time behind the voltage applied to the analog input 2.

Note For details of these parameters, refer to P.4-28 and 37 “Details of parameter”.

(2) Speed limit function

The speed limit is one of protective functions used during torque control.
This function regulates the motor speed so that it does not exceed the speed limit while the torque is controlled.

Caution

While the speed limit is used to control the motor, the torque command applied to the motor is not directly proportional to the analog torque command. Torque command should have the following result: the motor speed is equal to the speed limit.

• Relevant parameters <Selection of torque command 1, 3>

Parameter No.	Title	Range	Unit	Function
Pr3.21	Speed limit value 1	0 to 20000	r/min	Set up the speed limit used for torque controlling.
Pr3.22	Speed limit value 2	0 to 20000	r/min	
Pr3.15	Speed zero-clamp function selection	0 to 3	—	You can set up the function of the speed zero clamp input.

• Relevant parameters <Selection of torque command 2>

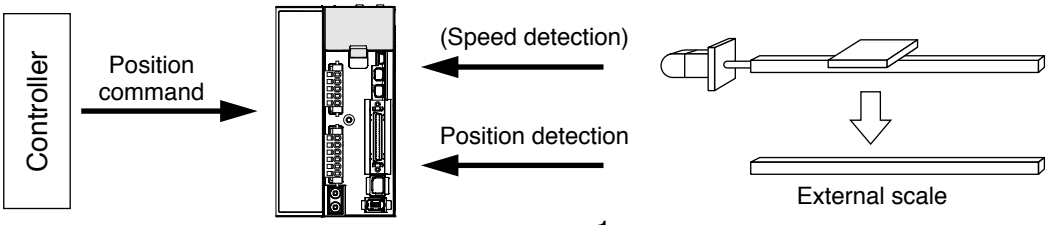
Parameter No.	Title	Range	Unit	Function
Pr3.02	Input gain of speed command	10 to 2000	(r/min) /V	Based on the voltage applied to the analog speed command (SPR), set up the conversion gain to motor command speed.
Pr4.22	Analog input 1 (AI1) offset setup	–5578 to 5578	0.359mV	Set up the offset correction value applied to the voltage fed to the analog input 1.
Pr4.23	Analog input 1 (AI1) filter	0 to 6400	0.01ms	Set up the time constant of 1st delay filter that determines the lag time behind the voltage applied to the analog input 1.
Pr3.15	Speed zero-clamp function selection	0 to 3	—	You can set up the function of the speed zero clamp input.

Note

For details of these parameters, refer to P.4-25, 27, 29 and 37 “Details of parameter”.

Full-closed Control Mode

In this full-closed control, you can make a position control by using a external scale mounted externally which detects the position directly and feeds it back. With this control, you can control without being affected by the positional variation due to the ball screw error or temperature and you can expect to achieve a very high precision positioning in sub-micron order.



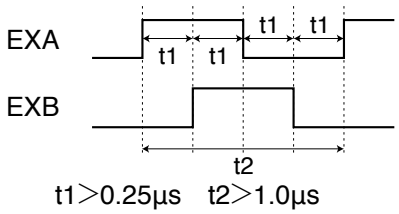
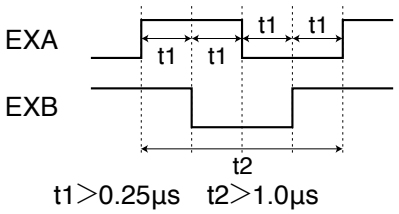
We recommend the external scale division ratio of $\frac{1}{40} \leq$ External scale division ratio ≤ 160

Cautions on Full-Closed Control

(1) Enter the command pulses making the external scale as a reference.
If the command pulses do not match to the external scale pulses, use the command division/multiplication function (Pr0.09 to Pr0.10) and setup so that the command pulses after division/multiplication is based on the external scale reference.

(2) The A5 series supports the external scale of A- and B-phase output type and serial communication type. Initialize the parameters according to the following procedures and write to EEPROM and turn on power.

(3) When using a scale of A- and B-phase output type, correctly connect it so that the rotating direction of the motor (CW/CCW) and A-phase and B-phase of the external scale have the following relationship.

CCW direction	CW direction
	
t1>0.25µs t2>1.0µs	t1>0.25µs t2>1.0µs

<How to make an initial setup of parameters related to external scale >

1) Turn on the power after checking the wiring.
2) Check the values (initial) feedback pulse sum and external scale feedback pulse sum with the front panel or with the setup support software, PANATERM.
3) Move the work and check the travel from the initial values of the above 2).
4) If the travel of the feedback sum and the external scale feedback pulse sum are reversed in positive and negative, set up the reversal of external scale direction (Pr3.26) to 1.
5) Set up the external scale division ratio (Pr3.24, Pr3.25) using the formula below,

$$\text{External scale division ratio} = \frac{\text{Total variation of feedback pulse sum}}{\text{Total variation of external scale feedback pulse sum}} = \frac{\text{Pr3.24}}{\text{Pr3.25}}$$

* If the design value of the external scale division ratio is obtained, set up this value.

Related page • P.3-17 “Control Block Diagram” • P.3-19 “Wiring Diagram to the connector, X4”
• P.3-30 “Inputs and outputs on connector X4” • P.4-3 “Details of parameter”

6) Set up appropriate value of hybrid deviation excess (Pr3.25) in 16 pulse unit of the external scale resolution, in order to avoid the damage to the machine.
   * A5-series driver calculates the difference between the encoder position and the external scale position as hybrid deviation, and is used to prevent the machine runaway or damage in case of the external scale breakdown or when the motor and the load is disconnected.
   If the hybrid deviation excess range is too wide, detection of the breakdown or the disconnection will be delayed and error detection effect will be lost. If this is too narrow, it may detect the normal distortion between the motor and the machine under normal operation as an error.
   * When the external scale division ration is not correct, hybrid deviation excess error (Err25) may occur especially when the work travels long distance, even though the external scale and the motor position matches.
   In this case, widen the hybrid deviation excess range by matching the external scale division ratio to the closest value.

Function

(1) Selection of external scale type

Select the type of external scale to be used.

• Relevant parameters

Parameter No.	Title	Range	Function
Pr3.23	External scale selection	0 to 2	Select the type of external scale.
Pr3.26	Reversal of direction of external scale	0 to 1	Reverse the direction of external scale, feedback counter.

Note For details of these parameters, refer to P.4-29 and 30 “Details of parameter”.

(2) Setup of external scale division ratio

Set up the division ratio of encoder resolution and external scale resolution.

• Relevant parameters

Parameter No.	Title	Range	Function
Pr3.24	Numerator of external scale division	0 to 2^{20}	Set up the numerator of the external scale dividing setup.
Pr3.25	Denominator of external scale division	1 to 2^{20}	Set up the Denominator of the external scale dividing setup.

Note For details of these parameters, refer to P.4-30 “Details of parameter”.

(3) Setup of hybrid excessive deviation

This function detects the positional difference between the motor (encoder) and load (external scale) and enables the hybrid excessive deviation protection if the difference exceeds Pr3.28 “Hybrid excessive deviation setup”.
Hybrid excessive deviation is mainly caused by feedback scale error, wrong connection and loose connection between the motor and load.

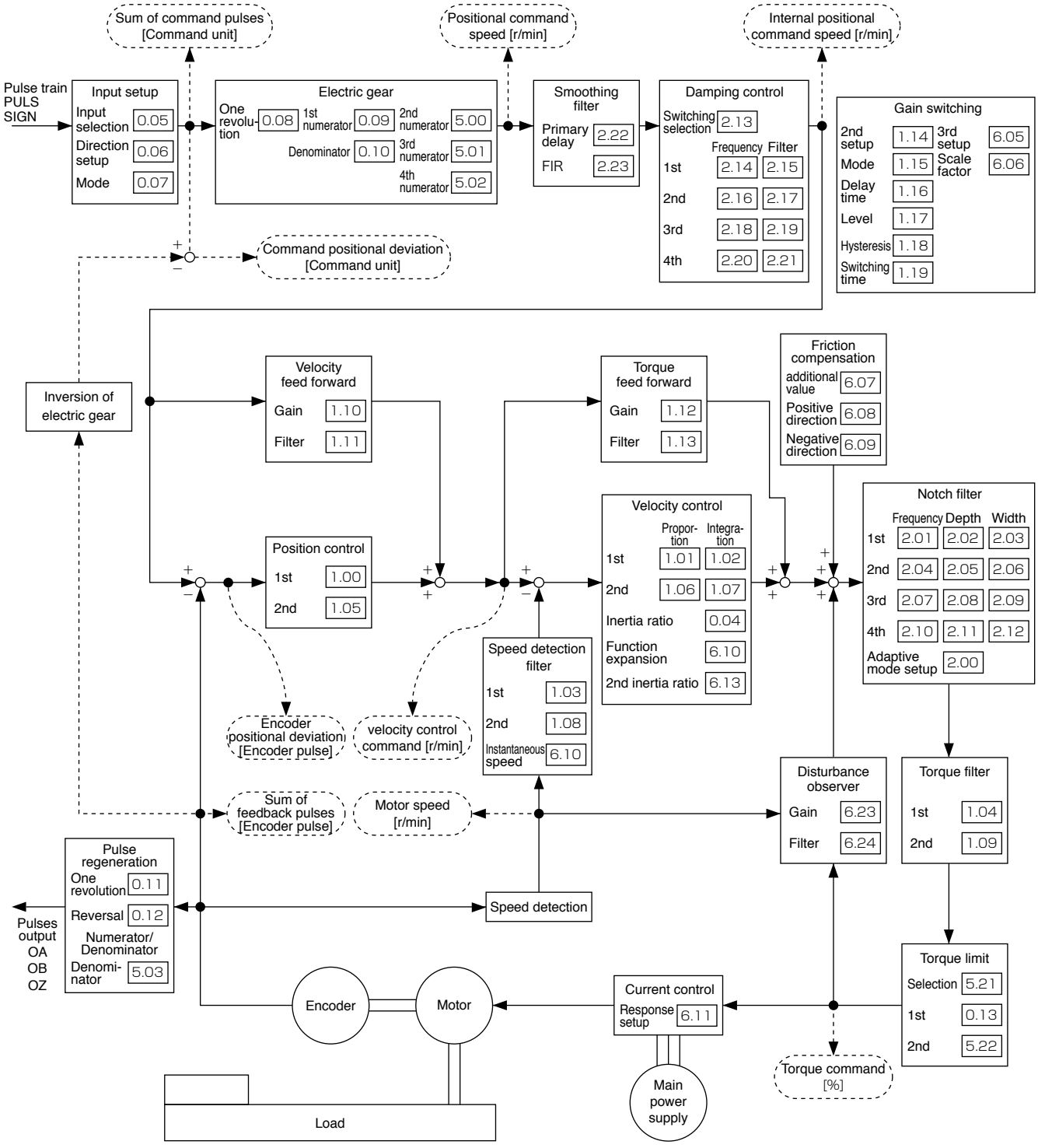
• Relevant parameters

Parameter No.	Title	Range	Function
Pr3.28	Hybrid deviation excess setup	1 to 2^{27}	You can setup the permissible gap (hybrid deviation) between the present motor position and the present external scale position.
Pr3.29	Hybrid deviation clear setup	0 to 100	As the motor turns the number of revolutions set by this parameter, the hybrid deviation is cleared to 0.

Note For details of these parameters, refer to P.4-31 “Details of parameter”.

2. Control Block Diagram

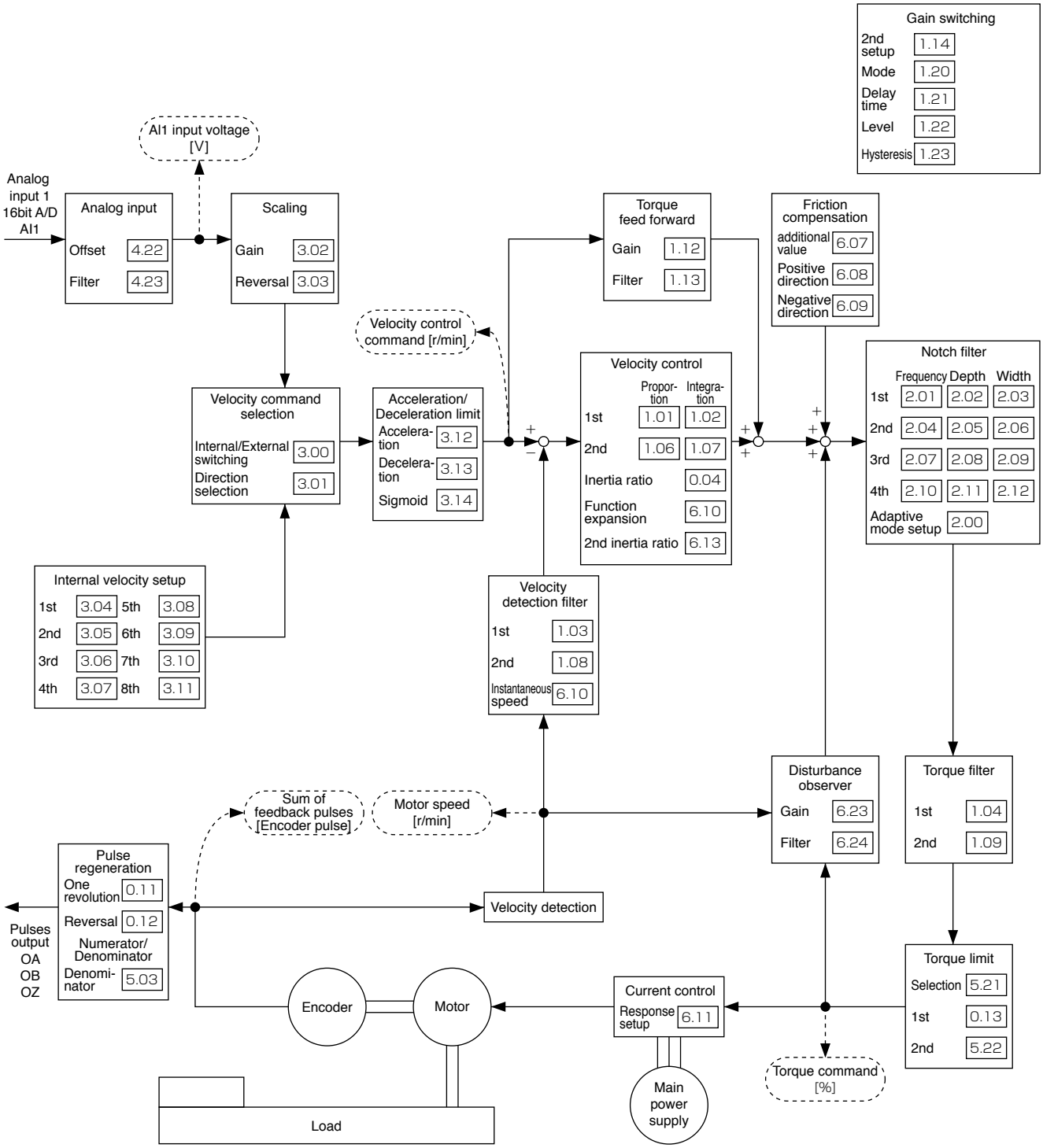
Position Control Mode



Related page
• P.3-18 “Wiring example of position control mode”
• P.3-20 “Connecting example to host controller”
• P.3-30 “Inputs and outputs on connector X4” • P.4-2 “List of Parameters”

2. Control Block Diagram

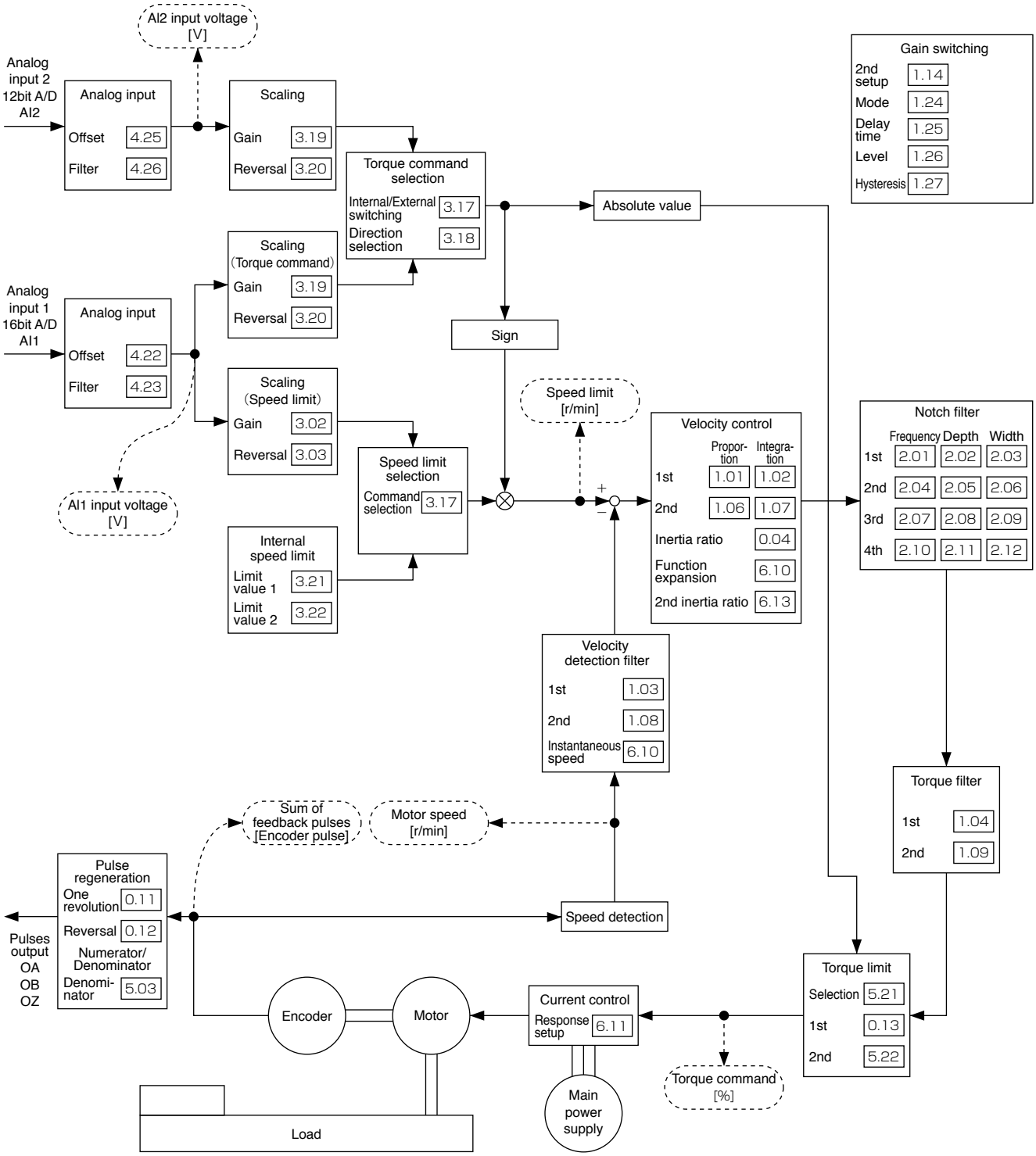
Velocity Control Mode



Related page • P.3-18 “Wiring example of velocity control mode”

2. Control Block Diagram

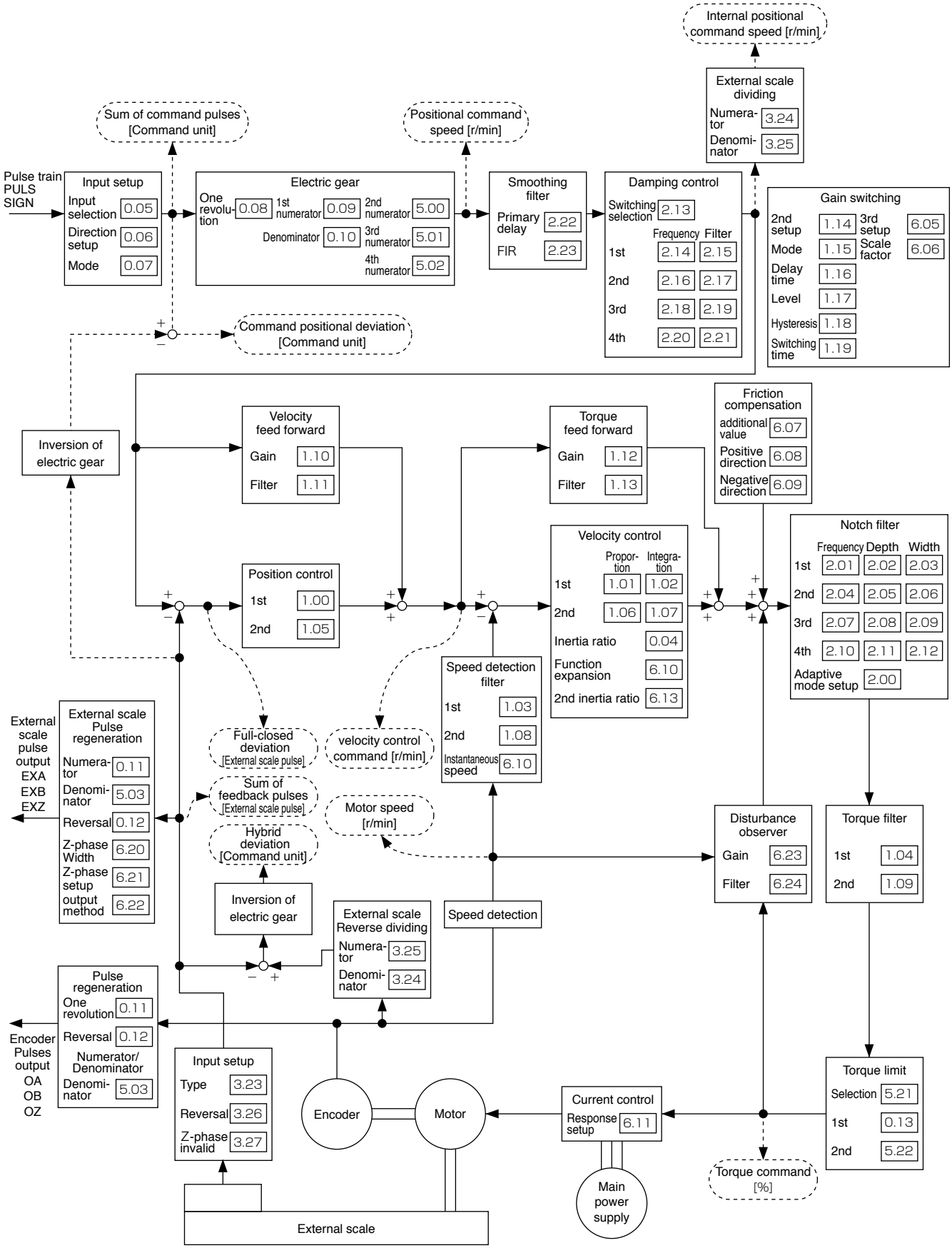
Torque Control Mode



Related page • P.3-19 “Wiring example of torque control mode”

2. Control Block Diagram

Full-closed Control Mode



Related page • P.3-19 “Wiring example of full-close control mode”